

LATEST PROGRESS ON TWO NEW ALS-U BEAMLINES FOR DIFFRACTION-LIMITED PERFORMANCE*

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Abstract

The Advanced Light Source Upgrade (ALS-U) will increase soft X-ray coherent flux by 100×. We developed two new beamlines, engineered to minimize loss of brightness and utilize the advanced coherence of the light source. Each beamline uses a minimalist optical layout: a cryogenically-cooled silicon M1 mirror, a monochromator with variable-line-spacing gratings, and a final focusing M3 mirror. Optics are designed for Strehl ratio > 0.8 and sub-100 nrad vibration. A piezo-bimorph M3 mirror paired with a wavefront sensor allows for wavefront optimization. Fabrication is underway. New test data include at-wavelength efficiency measurements for blazed gratings, and motion performance of piezo-actuated pitch/roll flexure systems at cryogenic temperatures, granite air-bearing positioners, and monochromators.

THE ALS-U PROJECT

The Advanced Light Source Upgrade (ALS-U) aims to increase the soft X-ray coherent flux by 100× at 1 keV [1]

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beyond the current ALS capabilities. The new storage ring will operate at 2 GeV with a current of 500 mA. It will deliver an electron beam size and divergence of about 10 μm RMS and 5 μrad RMS to the insertion device beamlines. The ALS-U project selected two beamlines for a complete rebuild.

The COherent Scattering and MICroscopy Beamline (COSMIC) will feature a single branch for coherent imaging. The brightness increase will enable 100× faster experiments, or 3× better spatial resolution, down to ms and nm.

The Microelectronic and Electronic STRucture Observatory (MAESTRO) will feature two branchlines for Angle-Resolved Photoemission Spectroscopy (ARPES). The higher coherent flux will boost the collection efficiency by 10× for better spatial resolution and faster measurements.

They have been engineered to minimize brightness loss and utilize the advanced coherence of the new source.

BEAMLINE DESIGN APPROACH

The beamline CAD model is shown in Fig. 1. The two beamlines share a simple layout with mutualized designs: a cryogenically-cooled silicon M1 mirror, a variable included angle plane grating monochromator (PGM) with variable-line-spacing (VLS) gratings, and a final-focusing plane-ellipse M3 mirror(s). The ALS-U scope ends at the exit slits and does not include the experimental endstations.

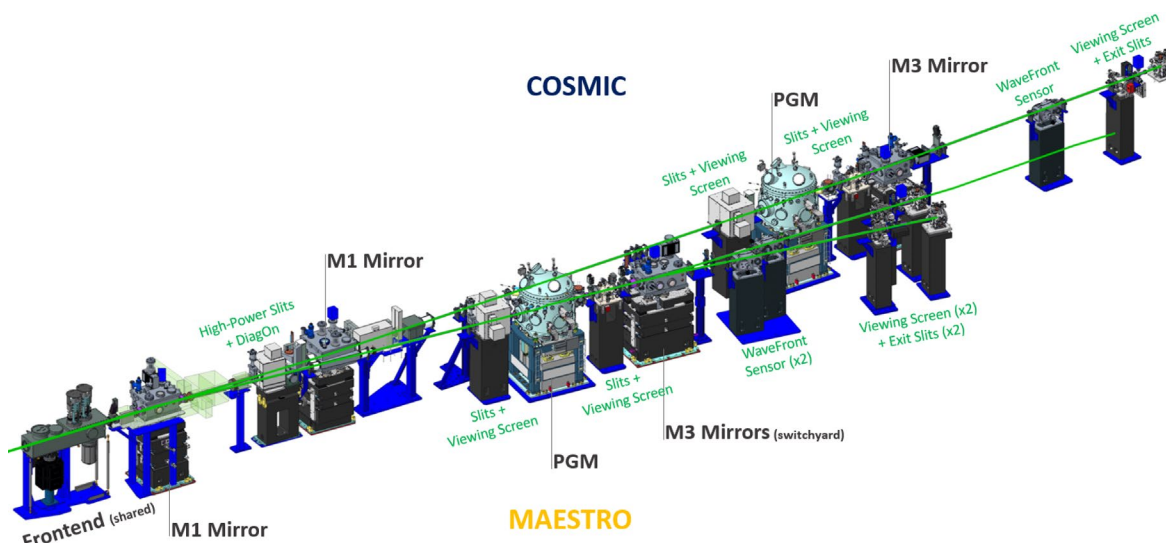


Figure 1: COSMIC and MAESTRO beamline layouts. Each beamline is approximately 30 m long and they share the same frontend. COSMIC has a single branchline, while MAESTRO splits at a M3 mirror switchyard. Optical elements are labelled, and diagnostics are highlighted in green.

Table 1 summarizes the performance goals for COSMIC and MAESTRO, both of which will reuse their existing elliptically polarizing undulator (EPU).

Table 1: Source and Beamline Specifications

	COSMIC	MAESTRO
EPU No of Poles	109	51
Peak Field (T)	0.759	1.098
Energy Range (eV)	250 - 2,500	60 - 600
Photon Energy Resolution	> 5,000	> 15,000
Beam Size at Exit Slit (μm)	10×10	25×18
Coherent Flux (ph/s)	6×10^{12} at 1 keV	1×10^{12} at 150 eV
Intensity stability	1% (0.1%*)	1% (0.1%*)

*best effort

To preserve the source's coherent flux and brightness, the beamline designs follow five design pillars, described in the following sections.

Thermal Management

The M1 mirror systems are designed to achieve a Strehl ratio of 0.8, which sets the requirement for the maximum allowable surface height error, see Fig. 2. To meet this requirement, we use a novel liquid-nitrogen (LN)-cooled cantilever M1 mirror design developed for ALS-U [2].

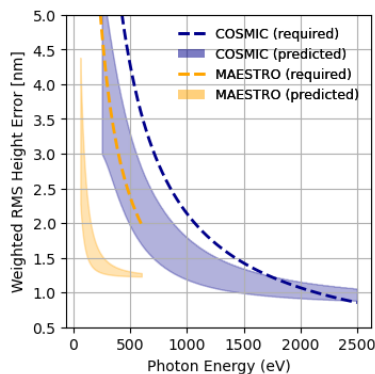


Figure 2: The maximum allowable height error is a function of photon energy, grazing angle, and Strehl ratio (0.8) and is therefore specific to each beamline (dashed lines). The shaded envelope shows the error budget: M1 thermal-structural deformation interpolated from a range of heat loads [2], plus fabrication errors from M1 and M3.

Originally designed for an absorbed heat load of 220 W and peak power density of 1 W/mm², the COSMIC and MAESTRO M1 absorb up to 80 W (0.2 W/mm²) and 150 W (0.5 W/mm²), respectively. We built a height error budget for the horizontal optics by adding thermal-structural deformation and fabrication (polishing) errors in quadrature: 0.7 nm RMS for M1, 0.4 nm RMS for the COSMIC M3, or 1.0 nm RMS for the MAESTRO M3.

The shaded envelope in Fig. 2 shows the range of thermal-structural deformation interpolated from [2]. The predicted height errors meet the requirement, except for COSMIC at the highest photon energies.

The PGMs consist of a vertically deflecting flat M2 mirror and VLS gratings (three for COSMIC, four for MAESTRO), all side-water-cooled. The height error tolerances for the M2 and gratings caused by thermal-structural deformation are more relaxed, at about 6 nm RMS for COSMIC and 2 nm RMS for MAESTRO, since adjustments to the VLS trajectories can compensate for quadratic deformation.

Low Vibration

Vibration requirements are set to keep intensity variation at the exit slits within 1%. That corresponds to about 100 nrad RMS vibration for each optic, with a best-effort target of 30 nrad RMS. Our approach is to design the horizontally deflecting optics, M1 and M3, with a First Natural Frequency (FNF) above 180 Hz. That requirement was validated in the 2024 test campaign of the mirror piezo-actuated pitch/roll flexure prototype [3].

For the PGMs, the M2 mirror and gratings share the same support structure, so in-phase motion (between optics) has little effect on beam stability. Instead, the first out-of-phase mode is specified above 80 Hz.

Wavefront Optimization

The beamlines use Ultra-High Vacuum (UHV) piezo-bimorph M3 mirrors [4] paired with a wavefront sensor to actively correct wavefront error in the horizontal direction [5]. The M3 systems use the same stiff architecture as the M1, with the uncooled bimorph mirror essentially replacing the LN-cooled cantilevered M1 mirror within a virtually identical assembly.

Carbon Contamination Mitigation

Carbon buildup can reduce soft X-ray flux by more than 80% at the carbon edge. It can also induce wavefront errors. To mitigate this, COSMIC and MAESTRO follow strict UHV design, fabrication, and testing protocols. Pumping is done by very high-capacity non-evaporable getter pumps. In addition, each optic features three in-situ contamination control methods: three switchable reflective stripes, oxygen cleaning, and hydrogen plasma cleaning.

Extensive Beam Diagnostics

Both beamlines include extensive photon diagnostics for fast, automated alignment. COSMIC features a DiagOn [6] with two multilayer mirrors at 22.5° incidence: one for invasive measurements and one with a notch for non-invasive monitoring, to enable source beam position monitoring before M1. Between each optic, diamond screens and slits provide beam monitoring and shaping. The exit slits offer sub-250 nm positioning precision and photocurrent readout, for fast beam steering (via pitch actuation of M3).

PRELIMINARY TEST RESULTS

Granite Positioner

The M1 and M3 mirror systems are positioned using granite air-bearing stages adapted from a design developed at the Advanced Photon Source [7]. They provide horizontal motion for coarse beam steering and vertical motion for switching reflective stripes.

Particular attention was given to yaw repeatability, especially for the switchyard M3 mirror. The granite positioners have been evaluated with a dummy load representing the mirror chamber (1000 kg) and applying lateral forces to simulate vacuum loads (70 kg), see Fig. 3.

Yaw runout was measured with a triple-head distance-measuring interferometer. The switchyard system demonstrated an angular repeatability of $2.4 \mu\text{rad}$.

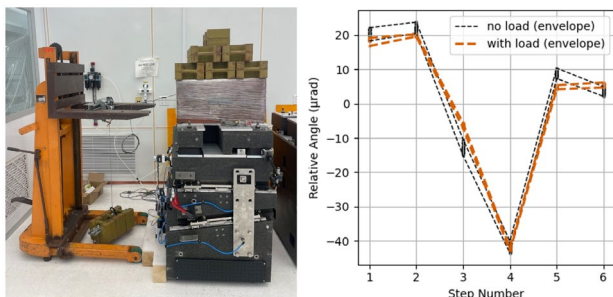


Figure 3: MAESTRO M3 granite positioner with dummy load (left). Yaw repeatability (right), evaluated over six positions across the full travel range and 20 cycles.

PGM

The PGMs were delivered and commissioned in May 2025. As part of the acceptance testing, we measured the frequency response of the MAESTRO PGM and extracted the modal parameters (frequency, shape, and damping). Instrumentation consisted of PCB Type 391A32 tri-axial accelerometers connected to the MEScope, and data processing with the Data Physics 900 series software. The PGM rested on four neoprene pads.

We excited a total of 60 nodes from two impact locations: on the granite stand and on the bearing towers, which support the in-vacuum mechanism and optics. We also performed “quiet” vibration measurements without impacts to determine the system’s transmissibility from the granite to the optics.

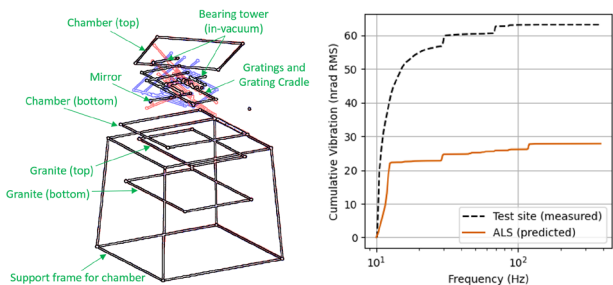


Figure 4: Extracted PGM's mode at 72 Hz (left). Mirror pitch vibration RMS > 10 Hz, measured at the test site and predicted at the beamline floor (right).

The first modes of interest for the grating and mirror (beyond rigid-body modes) appeared at 72 Hz, shown in Fig. 4, and 77 Hz. Using the accelerations from the final installation floor and the measured transmissibility, we estimated the pitch vibration of the mirror (taken as worst-case compared to the gratings) at 28 nrad RMS > 10 Hz.

Gratings

Six blazed gratings, designed for high throughput and all made by Inprentus, have been received. The COSMIC set also includes a lamellar grating to reduce harmonic contamination, fabricated by LBNL with a novel nanofabrication process [8]. Upon delivery, the grating's at-wavelength efficiencies are measured at ALS beamline 6.3.2. Figure 5 shows the results for five blazed gratings compared with their specifications.

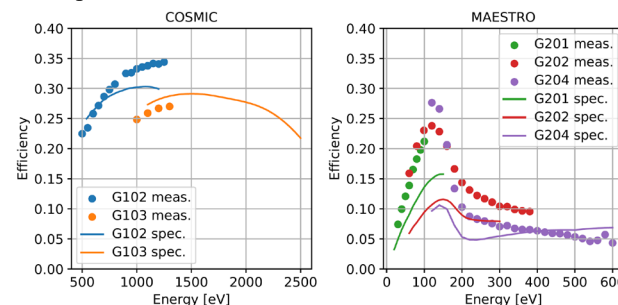


Figure 5: Measured at-wavelength efficiency compared with specifications: COSMIC (left), MAESTRO (right).

CONCLUSION

The ALS-U will increase soft X-ray coherent flux by 100×. COSMIC and MAESTRO are built to exploit this gain, with strict requirements for optical quality, thermal management, mechanical stability. Early fabrication and testing are encouraging. All beamline components are due within the next year, and the team is preparing installation.

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